

(TR-104) PYTHON WITH AI/ML

Training Day 41 Report

Date: 9 March 2026

Topic Covered: Generative Fine-Tuning of LLM for FAQ-Based Question Answering

The forty-first day of training focused on fine-tuning a lightweight Large Language Model for FAQ-based question answering using a custom dataset and transformer training workflow.

- Worked on generative fine-tuning of a language model using an FAQ dataset.
- Prepared and cleaned the dataset for training and evaluation.
- Converted dataset into CSV and JSONL formats for model training compatibility.
- Split dataset into training and testing sets for proper evaluation workflow.
- Set up TinyLlama-1.1B-Chat model and tokenizer for fine-tuning.
- Configured tokenization pipeline with maximum sequence length of 256 tokens.
- Processed and formatted data for transformer-based training.
- Fine-tuned the model for 3 training epochs.
- Saved the fine-tuned model for later inference and testing.
- Tested the fine-tuned model using sample prompts from the FAQ dataset.
- Verified response quality and checked whether generated answers matched expected outputs.
- Strengthened understanding of generative fine-tuning workflows for Large Language Models.
- Learned how custom datasets are prepared and formatted for instruction-style training.
- Explored transformer-based tokenization, sequence handling, and model training pipelines for conversational AI systems.
- **GitHub Link:**
<https://github.com/JaspinderKaurWalia26/projects/tree/main/src>

- **Blog Link:**

<https://medium.com/@jaspinderkaurwalia855/exploring-retrieval-techniques-in-semantic-search-basic-mmr-hybrid-c13c4628bc11>